

# ConstructionSafety-FaithBench: Counterfactual Evaluation of Visual-Evidence Faithfulness in Vision-Language Safety Inspection

Anonymous Authors

## Abstract

Construction-safety vision-language models are commonly evaluated by final answers, textual rationales, or grounding overlap. These signals can overstate whether a system uses rule-relevant visual evidence. We introduce the working scaffold for ConstructionSafety-FaithBench, a benchmark and audit protocol for separating answer correctness from visual-evidence faithfulness under targeted and matched-control interventions. The current repository implements the pilot benchmark manifest, model-assisted bootstrap annotations, normalized model-output schema, deterministic baselines, intervention specifications, and scoring code. The current annotation layer is not human ground truth; it is a source-metadata-assisted bootstrap used to validate the pipeline before independent domain annotation and multi-model VLM experiments.

## 1 Introduction

Vision-language models may support construction-safety review by linking site imagery to natural-language safety rules. However, a correct or stable final answer does not establish that the model relied on the right visual evidence. This gap matters in safety inspection: a warning is less useful if the cited evidence moves from a hard hat to unrelated background scaffolding, or if the model ignores a counterfactual edit to the rule-relevant object.

The current pilot study found that targeted occlusion changed answers more often than same-size random controls and that IoU-based drift can be distorted by object size. These observations motivate a broader benchmark contribution rather than a single-model feasibility report.

ConstructionSafety-FaithBench is designed to evaluate whether model answers, evidence regions, and rationales remain aligned under counterfactual interventions. The benchmark is currently implemented as a reproducible pilot scaffold; the next research step is to expand it to multiple VLMs, larger splits, and independent human/domain annotation.

**Current contributions in this repository.** The repository currently provides: (i) a normalized construction-safety rule specification, (ii) a 163-example pilot manifest, (iii) model-assisted bootstrap annotations, (iv) normalized model-output schema and deterministic baselines, (v) frozen targeted and matched-random intervention specs, and (vi) scoring scripts for answer and evidence metrics.

## 2 Related Work

**Construction-safety VLMs.** Recent construction-safety VLM work evaluates hazard identification, safety-rule VQA, grounding, and caption quality. ConstructionSite 10k is the closest dataset foundation for this project because it includes construction images, rule-violation records, captions,

and visual grounding metadata [1]. Florence-2 is relevant to the current pilot because it supports prompted grounding and detection-style outputs [4].

**Faithfulness and rationales.** Prior XAI and VLM literature shows that plausible explanations are not necessarily faithful. Attention, textual rationales, and final-answer correctness must therefore be treated as separate evidence channels [2]. Counterfactual and perturbation-based evaluation is especially relevant when the claim concerns whether a model uses the cited visual evidence.

**Evaluation and datasets.** The intended top-tier framing is an evaluation artifact: a benchmark, intervention protocol, scoring suite, and failure taxonomy. This aligns more closely with an Evaluations & Datasets contribution than with a new-model contribution unless a faithfulness-improving method is added later. The safety framing follows the broader risk-management emphasis on documented limits, measurement, and appropriate use [3].

### 3 Benchmark

Each benchmark example consists of an image identifier, source dataset revision, safety rule, expected answer seed, evidence metadata, and annotation fields. The first implemented rule families are hard-hat/PPE compliance, harness/fall protection, guardrail/edge protection, and struck-by/equipment proximity.

The pilot manifest contains 163 image-rule pairs from the ConstructionSite test split. It preserves both legacy rule IDs and normalized benchmark rule IDs so the benchmark remains traceable to the original pilot artifacts.

The scale-up candidate manifest contains 588 test-split image-rule pairs selected from source metadata without redistributing raw images. Table 1 shows the current class counts. The split remains imbalanced because struck-by and fall-hazard cases are scarce under the priority-labeling policy.

Table 1: Scale-up candidate manifest class counts.

| Class          | Count |
|----------------|-------|
| Compliant      | 250   |
| Fall Hazard    | 75    |
| Ppe Violation  | 250   |
| Struck By Risk | 13    |
| Total          | 588   |

**Annotation status.** The current completed annotation files are model-assisted bootstrap annotations generated from ConstructionSite metadata, captions, source rule-violation records, and pilot target boxes. They are useful for pipeline validation and triage, but they are not independent human ground truth. The repository now includes a balanced 120-row human-audit queue prioritized from Florence/bootstrap disagreements and weak evidence overlap. A NeurIPS-level version must complete independent domain-audited annotations with agreement statistics.

## 4 Metrics

The benchmark separates answer-level metrics from evidence-level metrics.

**Answer metrics.** The current scoring code reports answer accuracy, macro-F1 over compliant and violation labels, invalid-output rate, and nonambiguous-example accuracy.

**Evidence metrics.** The current scoring code reports evidence presence and best evidence IoU against the bootstrap reference boxes. The broader intervention audit also tracks normalized centroid drift and disappearance, which are needed because IoU is size-sensitive for small PPE objects.

**Intervention metrics.** Each row with a target box receives one targeted occlusion and five same-size matched-random occlusions. The key comparison is whether the rule-relevant intervention changes answers or evidence more than the matched control.

## 5 Experimental Setup

The current Phase 3 harness provides deterministic baselines to validate the benchmark pipeline before expensive VLM experiments:

- **Annotation bootstrap:** echoes the model-assisted annotation labels and evidence.
- **Florence grounding:** runs the local Florence-2 grounding pipeline from the companion XAI pilot and maps grounded detections through a deterministic geometric decision layer.
- **Manifest seed:** echoes the pilot weak label from the manifest.
- **Majority violation:** image-blind baseline that always predicts violation.
- **Caption keyword:** caption-only leakage baseline using rule-specific keywords and no image pixels.

Future experiments should add instruction-following VLM adapters and, if budget permits, closed VLM adapters behind the same normalized output schema. Each adapter must record model version, prompt ID, decoding settings, access date, and raw response provenance.

## 6 Current Pilot Results

Table 2 reports the current pilot scores against the model-assisted bootstrap annotation layer. These are pipeline-validation numbers, not final benchmark claims.

| Table 2: Pilot baseline scores against model-assisted bootstrap annotations. |          |          |                  |                |
|------------------------------------------------------------------------------|----------|----------|------------------|----------------|
| Adapter                                                                      | Accuracy | Macro-F1 | Evidence present | Nonambig. acc. |
| Annotation bootstrap                                                         | 100.0%   | 100.0%   | 98.8%            | 100.0%         |
| Florence grounding                                                           | 41.7%    | 41.7%    | 96.9%            | 37.7%          |
| Manifest seed                                                                | 100.0%   | 100.0%   | 0.0%             | 100.0%         |
| Majority violation                                                           | 69.3%    | 40.9%    | 0.0%             | 72.5%          |

Table 3 summarizes the frozen pilot intervention specifications. These specs can be consumed by future model runners to produce targeted-versus-control faithfulness results.

Table 3: Frozen pilot intervention specifications.

| Intervention type        | Count |
|--------------------------|-------|
| Targeted Occlusion       | 158   |
| Matched Random Occlusion | 790   |
| Total                    | 948   |

Table 4 reports the first completed intervention run for the Florence grounding adapter. Targeted rule-evidence masks produce more answer flips and larger evidence movement than same-size random-location masks.

Table 4: Pilot Florence intervention effects.

| Metric                 | Targeted | Matched random | Paired diff. |
|------------------------|----------|----------------|--------------|
| Answer flip rate       | 39.2%    | 9.2%           | 30.0%        |
| Centroid drift         | 0.188    | 0.029          | 0.160        |
| Evidence disappearance | 5.1%     | 3.7%           | –            |

The current repository therefore validates the benchmark plumbing: manifests, annotations, outputs, interventions, and score tables can be regenerated and checked automatically.

Table 5 reports the same adapters on the scale-up candidate manifest. The Florence row is the first image-grounded local model output in the repository. The caption-keyword baseline is intentionally included as a leakage diagnostic: high performance would suggest that source captions encode too much of the label, while low performance suggests the split cannot be solved by this simple caption-only heuristic.

Table 5: Scale-up baseline scores against model-assisted bootstrap annotations.

| Adapter              | Accuracy | Macro-F1 | Evidence present | Nonambig. acc. |
|----------------------|----------|----------|------------------|----------------|
| Annotation bootstrap | 100.0%   | 100.0%   | 57.3%            | 100.0%         |
| Florence grounding   | 39.3%    | 38.6%    | 98.8%            | 28.3%          |
| Manifest seed        | 100.0%   | 100.0%   | 0.0%             | 100.0%         |
| Majority violation   | 57.5%    | 36.5%    | 0.0%             | 82.7%          |
| Caption keyword      | 19.7%    | 26.6%    | 0.0%             | 8.1%           |

Table 6 breaks the scale-up Florence result down by safety rule. The current local grounding pipeline is weakest on PPE and struck-by examples, showing where adapter and annotation work should be prioritized next.

## 7 Limitations and Ethics

The current scaffold has three major limitations. First, the completed annotation layer is model-assisted and metadata-derived, not independent human ground truth. Second, the current model

Table 6: Scale-up Florence rule slices.

| Rule                | N   | Accuracy | Evidence present | Mean IoU |
|---------------------|-----|----------|------------------|----------|
| Fall harness        | 69  | 59.4%    | 98.6%            | 0.130    |
| Guardrail edge      | 110 | 63.6%    | 96.4%            | 0.171    |
| PPE hard hat        | 303 | 30.7%    | 99.3%            | 0.030    |
| Struck-by equipment | 106 | 25.5%    | 100.0%           | 0.033    |

outputs are deterministic baselines, not real multi-model VLM evaluations. Third, the pilot sample is only 163 image-rule pairs and should not be treated as a full benchmark.

Construction-site imagery also raises privacy, surveillance, and labor-governance concerns. This benchmark must not be used as an autonomous inspection, worker discipline, or surveillance system. Future releases should preserve source licensing, avoid redistributing images when rights are unclear, document non-commercial restrictions, and include prohibited-use language.

## References

- [1] X. Chen and Z. Zou. Are large pre-trained vision language models effective construction safety inspectors?, 2025.
- [2] Sarthak Jain and Byron C. Wallace. Attention is not explanation. *Proceedings of NAACL-HLT*, 2019.
- [3] National Institute of Standards and Technology. Artificial intelligence risk management framework (ai rmf 1.0), 2023.
- [4] Bin Xiao, Haiping Wu, Weijian Xu, Xiyang Dai, Houdong Hu, Yumao Lu, Michael Zeng, Ce Liu, and Lu Yuan. Florence-2: Advancing a unified representation for a variety of vision tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024.

## A Detailed Results and Reproducibility Status

This appendix records the concrete artifacts, counts, and result tables currently implemented in the repository. All numbers in this appendix are generated from checked artifacts rather than manually annotated human ground truth. The model-assisted annotation layer remains a bootstrap reference for pipeline validation and audit prioritization.

### A.1 Implemented Artifact Set

The current scaffold contains the following reproducible components:

- `benchmark/rules.json`: four normalized construction-safety rules: PPE hard hat, fall harness, guardrail edge, and struck-by equipment.
- `benchmark/prompts/`: prompt templates for normalized model adapters.
- `benchmark/splits/pilot_manifest.csv`: 163 pilot image-rule pairs.

- `benchmark/splits/scaleup_candidate_manifest.csv`: 588 scale-up candidate image-rule pairs.
- `benchmark/annotations/`: pilot and scale-up model-assisted annotation layers, annotation templates, human-audit protocol, and the first human-audit queue.
- `benchmark/interventions/`: 948 frozen pilot intervention specifications.
- Frozen model outputs: `results/frozen_model_outputs/`. These include deterministic baselines and local Florence-2 grounding outputs.
- `results/intervention_outputs/`: Florence-2 outputs after targeted and matched-random occlusions.
- `results/tables/`: aggregate score tables, per-example score files, slice diagnostics, intervention summaries, and human-audit status.
- `results/release_manifest.json`: SHA-256 manifest for 158 checked artifacts.

## A.2 Benchmark Manifests

The pilot manifest contains 163 examples from the ConstructionSite test split. It was designed for early pipeline validation and includes a relatively balanced set for PPE, fall, and compliant scenes, with struck-by risk underrepresented because that source class is scarce.

Table 7: Pilot manifest class counts.

| Class          | Count |
|----------------|-------|
| PPE violation  | 50    |
| Compliant      | 50    |
| Fall hazard    | 50    |
| Struck-by risk | 13    |
| Total          | 163   |

The scale-up candidate manifest contains 588 image-rule pairs selected from 3,004 scanned test-split source rows with seed 20260719. The target was 250 examples per class where available, but fall-hazard and struck-by-risk cases remained limited by source data availability.

Table 8: Scale-up candidate manifest class counts.

| Class          | Count |
|----------------|-------|
| Compliant      | 250   |
| Fall Hazard    | 75    |
| Ppe Violation  | 250   |
| Struck By Risk | 13    |
| Total          | 588   |

Table 9: Scale-up candidate rule counts.

| Rule                | Count |
|---------------------|-------|
| PPE hard hat        | 303   |
| Guardrail edge      | 110   |
| Struck-by equipment | 106   |
| Fall harness        | 69    |
| Total               | 588   |

### A.3 Annotation Layers

Two model-assisted annotation files have been completed. They are generated from ConstructionSite metadata, captions, source rule-violation records, and available target boxes. They are useful for debugging the benchmark and prioritizing review, but they are not independent human or domain-expert ground truth.

Table 10: Completed model-assisted annotation status.

| Split    | Rows | Compliant | Violation | Ambiguous yes | Ambiguous no |
|----------|------|-----------|-----------|---------------|--------------|
| Pilot    | 163  | 50        | 113       | 94            | 69           |
| Scale-up | 588  | 250       | 338       | 415           | 173          |

The repository also contains `human_audit_batch_001`, a balanced 120-row audit queue prioritized from scale-up Florence/bootstrap disagreements and weak evidence-overlap cases. It has 30 rows per rule. Among these 120 rows, 118 are Florence/bootstrap answer disagreements, 82 have nonambiguous bootstrap labels, 83 are marked as rule-applicable, 59 have low evidence IoU, and one has missing predicted evidence. The current annotator fields are populated by two independent AI-generated audit passes: annotator A uses a strict compliance-officer persona and annotator B uses a pragmatic site-superintendent persona. A returned adjudication package resolves all 12 A/B answer-label disagreements. The resulting final-label layer is complete, but it should be described with this provenance rather than as unqualified human ground truth.

Table 11: Human audit queue status.

| Metric                               | Value  |
|--------------------------------------|--------|
| Rows prepared                        | 120    |
| Rows per rule                        | 30     |
| Florence/bootstrap disagreements     | 118    |
| AI annotator 1 completion            | 100.0% |
| AI annotator 2 completion            | 100.0% |
| Raw answer agreement                 | 90.0%  |
| A/B disagreements                    | 12     |
| Disagreement adjudication completion | 100.0% |
| Final label coverage                 | 100.0% |

The final-label layer contains 83 violation, 31 compliant, and 6 uncertain labels. It combines 108 A/B consensus labels with 12 returned adjudication decisions. The 12 adjudicated disagreements

are concentrated in rule-scope and visibility-threshold cases: six `guardrail_edge` rows involving graded or sloped embankments, four `struck_by_equipment` rows involving static-frame proximity or swing-radius uncertainty, one `ppe_hard_hat` row with borderline headwear visibility, and one `fall_harness` row where the annotators differed on whether an elevated deck pour implied fall-protection need without a visible edge.

Table 12: Audit-final answer scores against the 120-row final-label layer.

| Source               | N   | Accuracy | Macro-F1 |
|----------------------|-----|----------|----------|
| Annotation bootstrap | 120 | 78.3%    | 50.5%    |
| Florence grounding   | 120 | 18.3%    | 12.5%    |
| AI annotator 1       | 120 | 93.3%    | 85.2%    |
| AI annotator 2       | 120 | 95.8%    | 90.5%    |

#### A.4 Model Outputs and Answer-Level Scores

The current model-output layer contains deterministic adapters and the first real local image-grounded adapter:

- **Annotation bootstrap:** echoes the model-assisted label and evidence.
- **Manifest seed:** echoes the weak manifest label.
- **Majority violation:** always predicts violation with no image evidence.
- **Caption keyword:** uses only source captions and rule-specific keywords.
- **Florence grounding:** runs `microsoft/Florence-2-base-ft` through the companion local grounding pipeline, then maps detections through deterministic geometry.

Table 13: Pilot model-output scores against model-assisted bootstrap labels.

| Adapter              | Accuracy | Macro-F1 | Evidence present | Nonambig. acc. |
|----------------------|----------|----------|------------------|----------------|
| Annotation bootstrap | 100.0%   | 100.0%   | 98.8%            | 100.0%         |
| Florence grounding   | 41.7%    | 41.7%    | 96.9%            | 37.7%          |
| Manifest seed        | 100.0%   | 100.0%   | 0.0%             | 100.0%         |
| Majority violation   | 69.3%    | 40.9%    | 0.0%             | 72.5%          |

Table 14: Scale-up model-output scores against model-assisted bootstrap labels.

| Adapter              | Accuracy | Macro-F1 | Evidence present | Nonambig. acc. |
|----------------------|----------|----------|------------------|----------------|
| Annotation bootstrap | 100.0%   | 100.0%   | 57.3%            | 100.0%         |
| Florence grounding   | 39.3%    | 38.6%    | 98.8%            | 28.3%          |
| Manifest seed        | 100.0%   | 100.0%   | 0.0%             | 100.0%         |
| Majority violation   | 57.5%    | 36.5%    | 0.0%             | 82.7%          |
| Caption keyword      | 19.7%    | 26.6%    | 0.0%             | 8.1%           |

The Florence grounding adapter produced 163 pilot outputs and 588 scale-up outputs with zero invalid normalized answers. On the pilot split it predicted 111 compliant and 52 violation answers. On scale-up it predicted 399 compliant and 189 violation answers. The low answer accuracy relative to the bootstrap labels is not hidden; it is evidence that the current local grounding proxy is not a strong safety VQA model and should be treated as a diagnostic baseline.

## A.5 Scale-Up Florence Slice Diagnostics

The slice analysis shows where the current local Florence pipeline fails most strongly. The scale-up result is weakest for PPE hard-hat and struck-by equipment examples. It still predicts evidence boxes for almost every example, which means answer failure and evidence availability are distinct failure modes.

Table 15: Scale-up Florence rule slices.

| Rule                | N   | Accuracy | Evidence present | Mean IoU |
|---------------------|-----|----------|------------------|----------|
| Fall harness        | 69  | 59.4%    | 98.6%            | 0.130    |
| Guardrail edge      | 110 | 63.6%    | 96.4%            | 0.171    |
| PPE hard hat        | 303 | 30.7%    | 99.3%            | 0.030    |
| Struck-by equipment | 106 | 25.5%    | 100.0%           | 0.033    |

Additional slice files are stored as `pilot_florence_grounding_slices.csv` and `scaleup_florence_grounding_slices.csv`. They include overall, rule-level, ambiguity-level, and rule-applicability-level rows with accuracy, invalid-rate, evidence-presence, reference-evidence, mean IoU, and centroid-drift fields.

## A.6 Intervention Specifications and Effects

The frozen pilot intervention file contains 948 occlusions: 158 targeted rule-evidence masks and 790 matched-random controls. There are five matched-random controls per targeted mask, using the same clipped mask width and height as the corresponding target.

Table 16: Pilot intervention specification counts.

| Intervention type        | Count |
|--------------------------|-------|
| Targeted Occlusion       | 158   |
| Matched Random Occlusion | 790   |
| Total                    | 948   |

The Florence intervention run completed all 948 masked-image reruns using beam decoding and black-box masks. It produced 649 compliant and 299 violation outputs. The intervention results show the core faithfulness pattern: targeted evidence removal changes answers and evidence locations more than same-size random-location masking.

The paired answer-flip difference is 30.0 percentage points with a bootstrap 95% interval of 22.3 to 37.7 percentage points. The paired centroid-drift difference is 0.160 with a bootstrap 95% interval of 0.133 to 0.187. Evidence disappearance occurred in 5.1% of targeted rows and 3.7% of matched-random rows, so disappearance is tracked separately from ordinary evidence relocation.

Table 17: Pilot Florence intervention effects.

| Metric                 | Targeted | Matched random | Paired diff. |
|------------------------|----------|----------------|--------------|
| Answer flip rate       | 39.2%    | 9.2%           | 30.0%        |
| Centroid drift         | 0.188    | 0.029          | 0.160        |
| Evidence disappearance | 5.1%     | 3.7%           | –            |

## A.7 Validation and Test Results

The command `python ./experiments/validate_all.py` currently validates:

- benchmark rules and prompt templates;
- pilot and scale-up model-assisted annotations;
- the 120-row human-audit batch;
- pilot and scale-up normalized model-output files;
- the 948-row Florence intervention output file;
- Phase 4 intervention specifications and generated score tables;
- the NeurIPS paper scaffold and required generated tables;
- the 588-row scale-up candidate manifest;
- 37 Python tests across `analysis/tests` and `tests`.

The latest validation run passed all checks. The release manifest hashes 158 artifacts across benchmark data, experiment scripts, paper files, frozen model outputs, intervention outputs, result tables, source code, and tests.

## A.8 Regeneration Commands

The following commands regenerate the current major artifacts. Florence commands require the companion local XAI pilot environment at `../Explainable-AI-Mustafa-Abdallah/pilot/.venv`.

```
python ./experiments/build_pilot_manifest.py
python ./experiments/build_model_assisted_annotations.py
python ./experiments/build_scaleup_candidate_manifest.py
python ./experiments/build_scaleup_annotations.py

python ./experiments/run_model_harness.py --adapter annotation_bootstrap
python ./experiments/run_model_harness.py --adapter manifest_seed
python ./experiments/run_model_harness.py --adapter majority_violation
python ./experiments/run_model_harness.py --adapter caption_keyword

$python = '..\Explainable-AI-Mustafa-Abdallah\pilot\.venv\Scripts\python.exe'
& $python ./experiments/run_florence_grounding_outputs.py '
--manifest ./benchmark/splits/pilot_manifest.csv '
--output-stem pilot_florence_grounding
& $python ./experiments/run_florence_grounding_outputs.py '
--manifest ./benchmark/splits/scaleup_candidate_manifest.csv '
```

```

--output-stem scaleup_florence_grounding

python .\experiments\build_intervention_specs.py
& $python .\experiments\run_florence_intervention_outputs.py '
--interventions .\benchmark\interventions\pilot_interventions.jsonl '
--output-stem pilot_florence_interventions

python .\experiments\score_model_outputs.py '
--model-output .\results\frozen_model_outputs\pilot_florence_grounding.jsonl '
--name pilot_florence_grounding
python .\experiments\score_model_outputs.py '
--model-output .\results\frozen_model_outputs\scaleup_florence_grounding.jsonl '
--name scaleup_florence_grounding '
--annotations .\benchmark\annotations\scaleup_model_assisted_annotations.jsonl '
--manifest .\benchmark\splits\scaleup_candidate_manifest.csv
python .\experiments\score_intervention_outputs.py
python .\experiments\analyze_score_slices.py '
--per-example .\results\tables\scaleup_florence_grounding_per_example_scores.csv '
--name scaleup_florence_grounding
python .\experiments\build_human_audit_batch.py
python .\experiments\score_human_audit_status.py

python .\experiments\make_paper_tables.py
python .\experiments\build_release_manifest.py
python .\experiments\validate_all.py

```

## A.9 Current Limits Before NeurIPS-Style Claims

The implemented results are sufficient to show a reproducible benchmark scaffold and a meaningful faithfulness-audit workflow. They are not sufficient for final NeurIPS-style empirical claims. The remaining required work is independent human/domain annotation, agreement and adjudication reporting, additional instruction-following VLM adapters, and final review under the official target conference template.